VISHWAKARMA INSTITUTE OF TECHNOLOGY, PUNE

Department of Artificial Intelligence and Machine Learning

FEW-SHOT TEMPORAL ANOMALY GENERATION AND DETECTION IN BRAIN CONNECTIVITY SUBGRAPHS FOR AUTISM DISORDER

Branch: AIML-B

GRN: 4

Day: Tuesday

Date: 13/10/2026

Project Guide

Dr. Premanand Ghadekar

Group Members

Sr. No.	Roll No.	Name	PRN	Email ID
1	58	Hanil Shah	12414322	hanil.shah24@vit.edu
2	59	Harsh Patel	12413448	patel.harsh24@vit.edu
3	62	Tanaya Hartalekar	12410318	tanaya.hartalekar24@vit.edu
4	73	Shravani Dixit	12415111	shravani.dixit24@vit.edu

INTRODUCTION

The Problem

- Autism (ASD) is linked to differences in brain connectivity seen in resting-state fMRI.
- Brain networks change over time, so one static network per scan can miss short-lived changes.
- Labelled ASD scans are few, so we learn what is normal from control scans and flag what deviates.
- We also create synthetic anomalies to see if they help when only a few ASD scans exist.

Project Framing

Can unusual changes in small brain subgraphs be found by learning only normal dynamics, and can synthetic anomalies help with few ASD examples?

Scope note: this is a research prototype for anomaly generation and detection, not a clinical diagnostic tool. No claim of ASD diagnosis is made.

OBJECTIVES

1 **Build time-resolved brain graphs**

Convert 871 ABIDE I scans (360 regions) into sliding-window brain graphs.

✓ COMPLETED

2 **Track brain-region subgraphs**

Follow 20 selected regions and their neighbours across time windows.

✓ COMPLETED

3 **Learn normal brain dynamics**

Train a GIN + GRU on control scans only to predict the next window.

✓ COMPLETED

4 **Detect anomalies**

Score prediction errors and set thresholds from validation controls.

✓ COMPLETED

5 **Generate synthetic anomalies**

Gradient-guided edge edits on control subgraphs (7,200 records, 0 rule violations).

✓ COMPLETED

6 **Evaluate few-shot and severity**

Test 1, 5, 10 ASD scans (accuracy, F1, AUROC); severity and demo app.

✓ COMPLETED

LITERATURE SURVEY

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Theme	Key Paper	Key Insight	Relevance to Project
ABIDE dataset	A. Di Martino et al. (2014)	Openly shares rs-fMRI + phenotypic data for 539 ASD / 573 control subjects across 17 sites; reports baseline hypoconnectivity in ASD.	Grounds the ABIDE I dataset (871 subjects, 20 sites) used throughout.
GNN expressive power	Keyulu Xu et al. (2018)	Proves common GNN variants are weaker than the Weisfeiler-Lehman test; introduces GIN, which matches it.	Justifies the weighted GIN encoder used to embed each subgraph.
Dynamic functional connectivity	R. Matthew Hutchison et al. (2013)	Defines and surveys sliding-window estimation of time-varying connectivity and its methodological pitfalls.	Grounds the 30-TR / stride-10 sliding-window graph construction.
Graph anomaly detection	Xiaoxiao Ma et al. (2021)	Taxonomizes deep graph-anomaly-detection methods by anomaly type and technique; flags scarce labels as an open challenge.	Frames the prediction-error, threshold-calibrated anomaly score.
Gradient-based graph perturbation	Kaidi Xu et al. (2019)	Formulates graph structure attacks as constrained optimization solved with projected gradient descent (PGD).	Grounds the PGD-style iterative edge-weight perturbation.

Existing Work → Our Focus



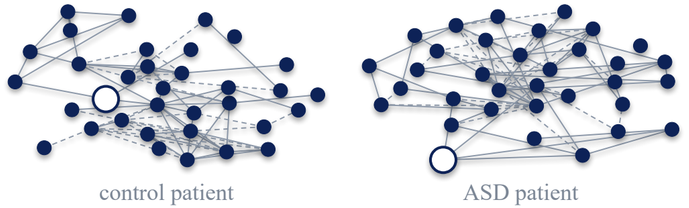
Our focus: existing work provides graph representations, temporal modeling, classification, and anomaly detection; this project additionally investigates structure-aware, gradient-guided anomaly generation for a few-shot brain-subgraph setting. Full 20-paper analysis: *GRAPH_ANOMALY_Literature_Review.xlsx*

PROPOSED METHODOLOGY

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DATA PREPARATION AND PATIENT SPLIT

ABIDE I, 871 patient scans



control patient ASD patient

SLIDING-WINDOW BRAIN GRAPH BUILDING

30-TR windows, stride 10; Pearson correlation; signed graph, top 20% edges

BRAIN-REGION SUBGRAPH TRACKING

20 selected brain regions (ROIs), 1-hop neighbourhood; node set fixed at the first window

GIN ENCODING AND GRU NEXT-WINDOW PREDICTION

2-layer weighted GIN, mean pool, 128-D embedding; GRU (hidden 128), MSE + variance loss, controls only

TEMPORAL SUBGRAPH ANOMALY SCORING FOR ASD DETECTION

patient score = max over regions and transitions

CONTROL-BASED DETECTION THRESHOLD CALIBRATION

95th percentile of validation-control scores (transition and patient level)

SYNTHETIC SUBGRAPH ANOMALY GENERATION

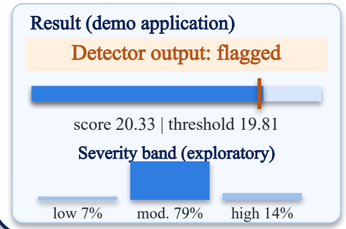
Gradient-guided PGD edge-weight edits, 5 steps, up to 15% of edge pairs, 3 severities

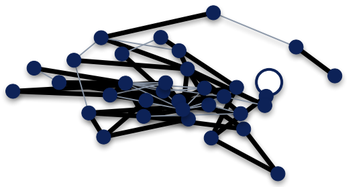
FEW-SHOT ASD ADAPTATION

1 / 5 / 10 ASD patient scans + 30 generated anomalies; logistic regression on 20-D region scores

ANOMALY DETECTION AND SEVERITY OUTPUT

Flag if score exceeds threshold; ADOS-2 severity band





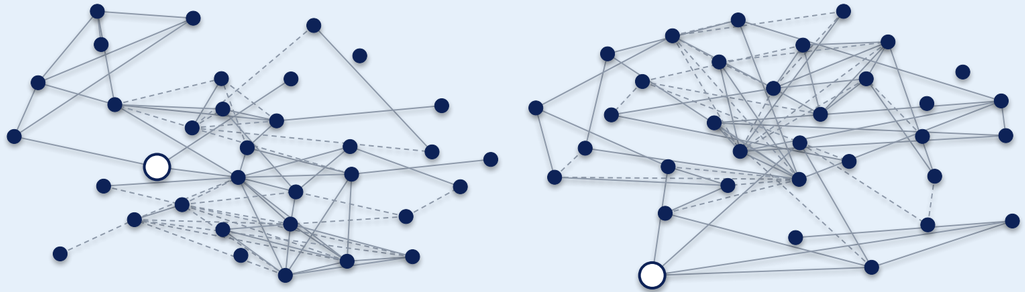
synthetic anomaly (bold = edited)

MAJOR CONTRIBUTIONS

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1 Tracking brain-region subgraphs over time

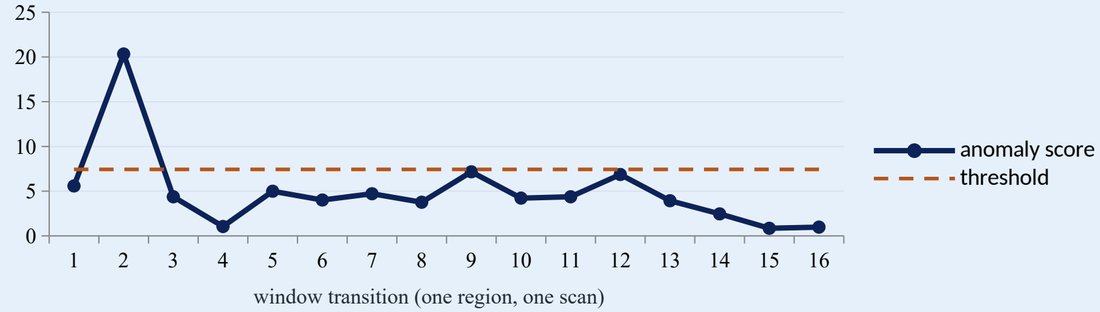
Small subgraphs are followed across time windows, so a change is located in region and time.



control patient ASD patient

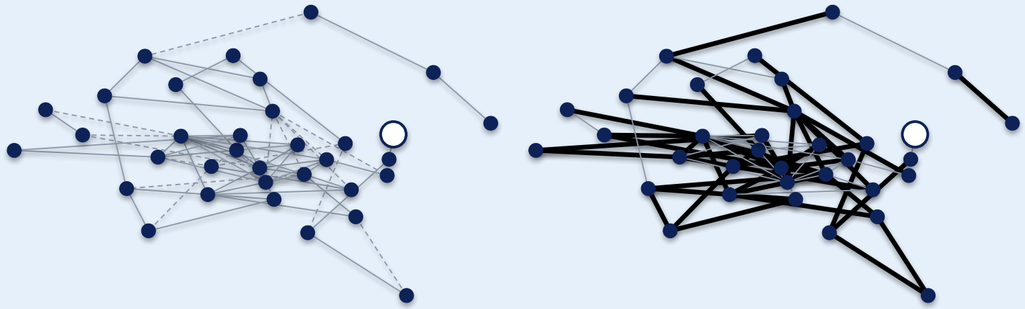
2 Control-only temporal anomaly scoring

A model learns normal dynamics from controls; the next-step prediction error is the anomaly score.



3 Gradient-guided anomaly generation

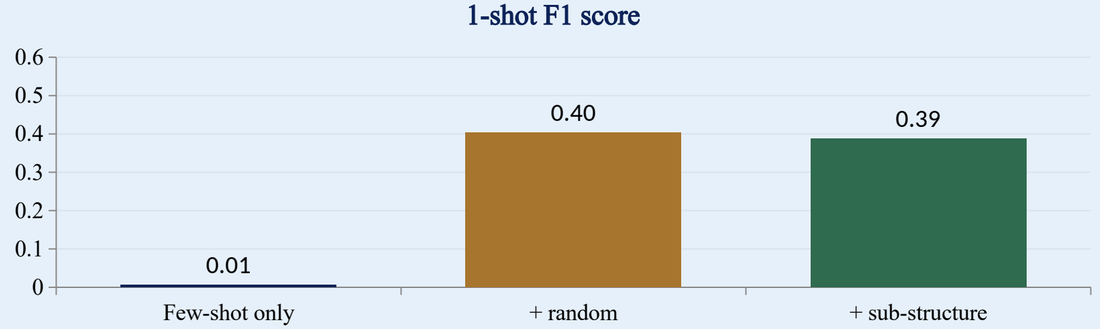
Weight-only edits of real control subgraphs create labelled anomalies: 45.8% detected vs 4.8% for random edits.



real control after guided edit (bold)

4 Few-shot adaptation with generated anomalies

A few ASD scans plus generated anomalies train the detector; 1-shot F1 rises from 0.006 to about 0.40.



RESULTS: OVERALL COMPARISON

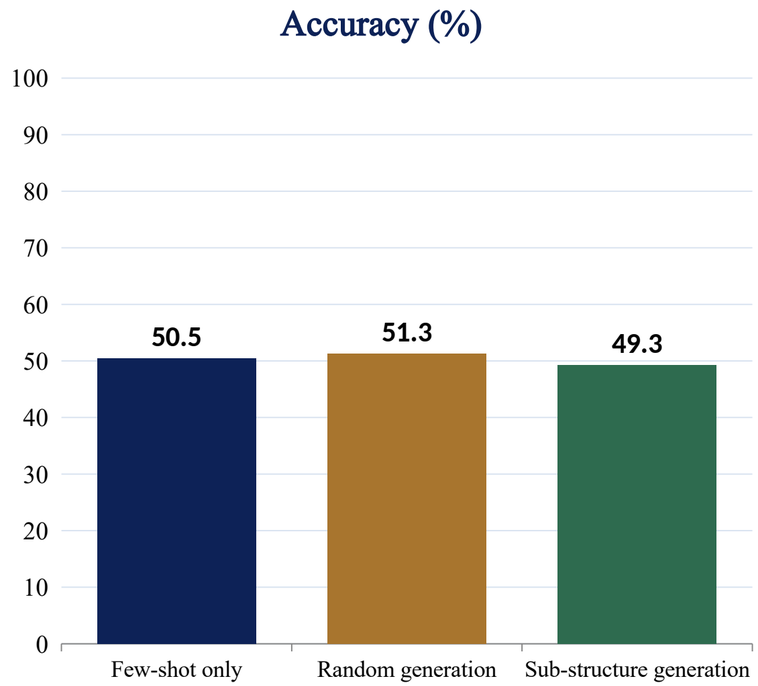
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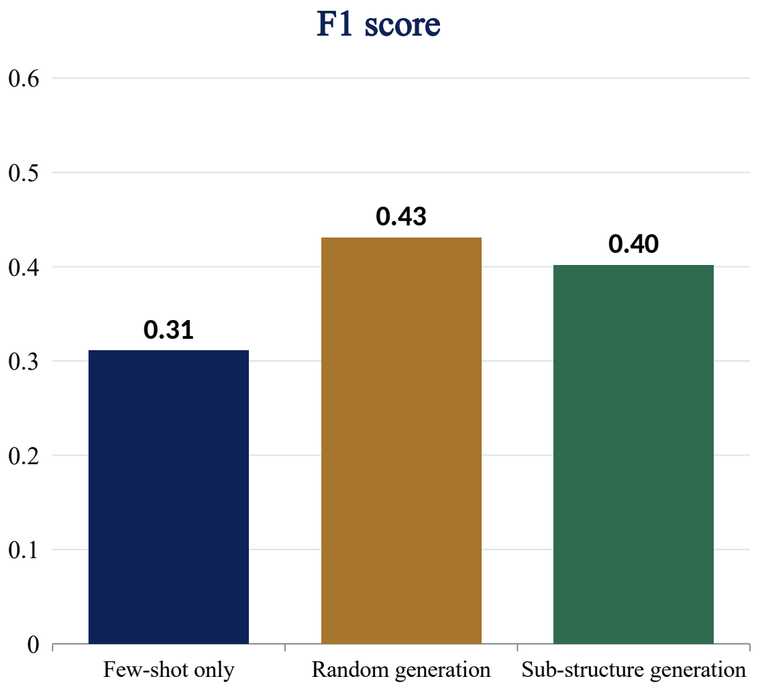
Approach	Accuracy	Precision	Recall	F1 score	AUROC
Baseline detector (no few-shot data)	53.4%	0.500	0.033	0.062	0.453
Few-shot data only (normal detection)	**52.1% ± 1.8**	0.354 ± 0.20	0.115 ± 0.05	0.155 ± 0.06	**0.508 ± 0.04**
Random anomaly generation + detection	51.0% ± 2.1	**0.466 ± 0.03**	**0.373 ± 0.07**	**0.411 ± 0.05**	0.507 ± 0.04
Sub-structure anomaly generation + detection	49.5% ± 2.6	0.445 ± 0.04	0.353 ± 0.05	0.393 ± 0.04	0.495 ± 0.04

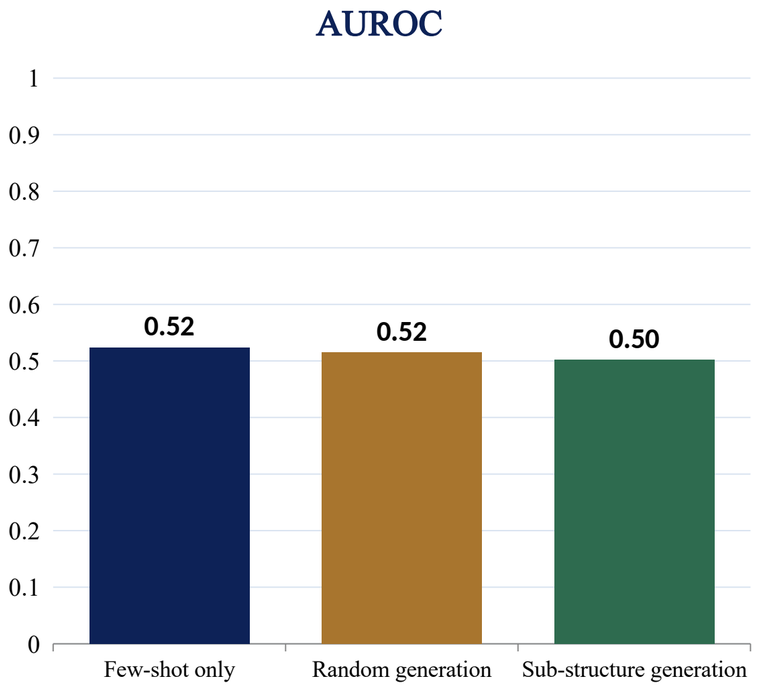
- **How to read:** average over 1, 5 and 10 ASD scans (each from five random draws), tested on 131 held-out patients; green marks the best value in each column.
- Adding generated anomalies raises recall and F1 over few-shot data alone.
- Accuracy and AUROC stay close to chance (about 50%) for every approach.

RESULTS: 10-SHOT COMPARISON

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9.6×

Sub-structure edits are detected 9.6× more often than random edits (45.8% vs 4.8%).

0.31 → 0.43

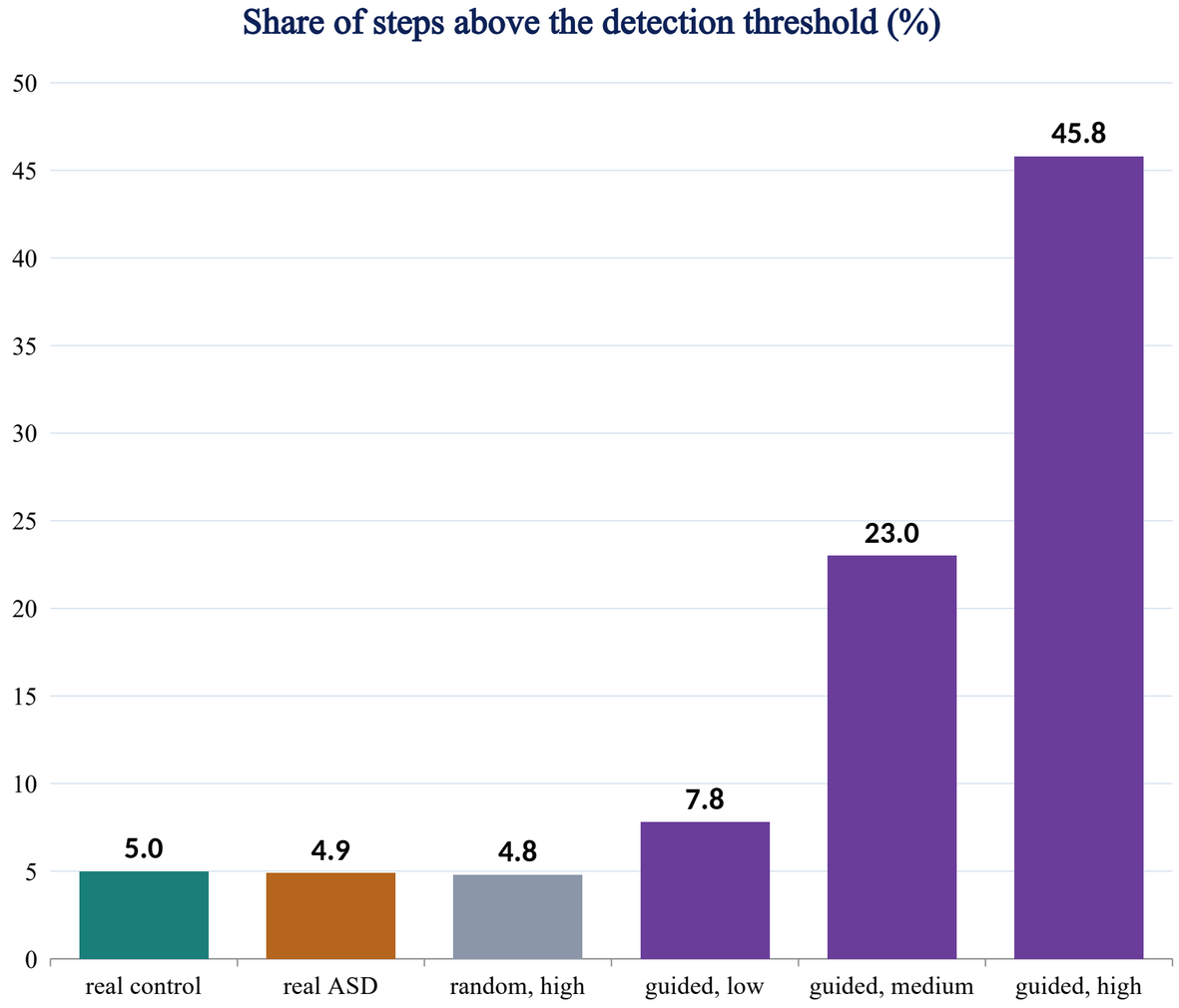
10-shot F1 rises from 0.31 with few-shot data alone to 0.43 (random) and 0.40 (sub-structure).

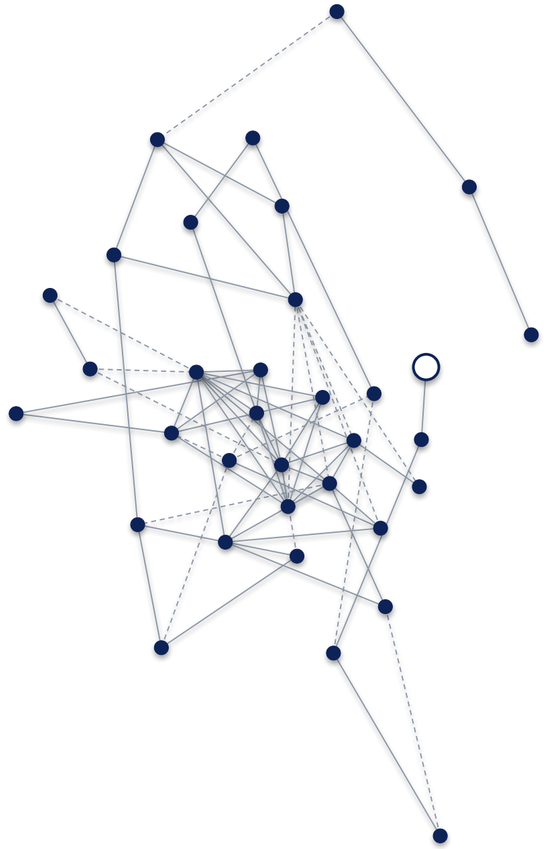
≈ 0.50

AUROC stays near chance for all three approaches: the gain comes from flagging more patients.

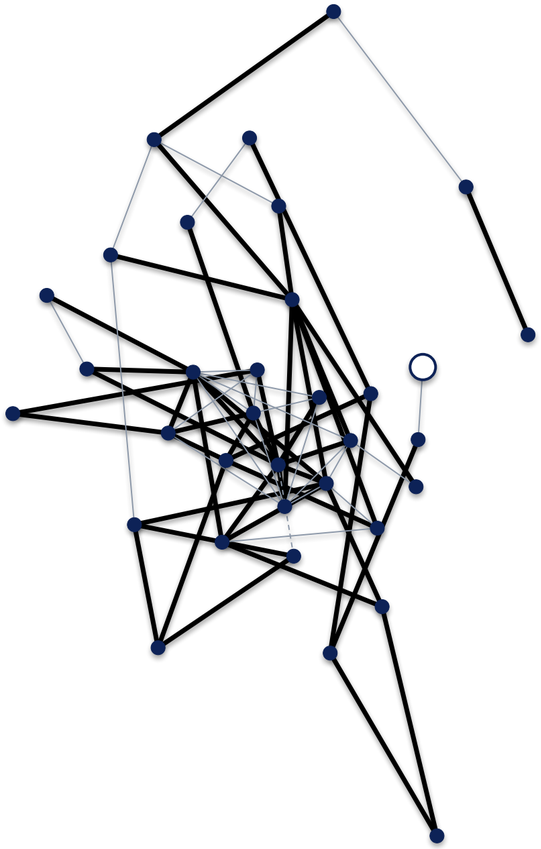
RESULTS: SYNTHETIC ANOMALIES

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real control subgraph



after guided edit (edited edges bold)

Guided edits: score 0.62 → 4.83 on this example; 45.8% of guided high-severity steps are detected vs 4.8% for random edits.

DISCUSSION

WHAT WORKED

- Stable training with no representation collapse (embedding std 1.03–1.16).
- Generator works as designed: 7,200 records, 0 rule violations, severity-monotonic.
- Sub-structure edits detected 45.8% of the time vs 4.8% for random edits (9.6×).
- Leakage-safe protocol: patient-level split, thresholds from validation controls only.
- Complete pipeline plus a demo application and 20 sample scans.

CHALLENGES

- Detector is at chance on ABIDE I: AUROC 0.453, accuracy 53.4%.
- A single step-level threshold flagged all 131 test patients; fixed with a patient-level threshold.
- Few-shot F1 gains come from flagging more patients; AUROC stays about 0.50.
- Severity estimate is below a majority guess (43.3% vs 55.0%).
- Only 20 of 360 regions, one training run, no site-level test.

INTERPRETATION

- A diagnosis label is a weak sign of unusual connectivity dynamics; ASD is highly varied.
- Site effects and short, noisy windows may hide a real signal.
- Detectability of generated anomalies is self-referential: the same model guides and scores the edits.
- Guided and random generation behave alike in the few-shot test.
- Research prototype: no clinical claim.

CONCLUSION

- Built and tested an end-to-end pipeline: time-resolved brain graphs, brain-region subgraph tracking, temporal anomaly scoring, synthetic anomaly generation, few-shot adaptation and a severity output.
- All six objectives were completed, including a working demo web application.
- The anomaly generator works as designed: 0 rule violations, and 45.8% of sub-structure edits are detected against 4.8% for random edits.
- On ABIDE I the detector is at chance (AUROC 0.453, accuracy 53.4%); few-shot F1 gains come from a higher flag rate (AUROC about 0.50).
- Severity estimation is below a majority guess (43.3% vs 55.0%).
- Research prototype only: no clinical claim, and all results are reported as found.

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